

Skills Summary

Evaluating Absolute-Value and Piecewise Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Evaluating an absolute-value rule

What the bars do: $|n|$ is the distance from n to zero, so it is never negative: $|5| = 5$ and $|-5| = 5$.

Order of work, every time: (1) finish everything *inside* the bars, (2) take the absolute value, (3) do whatever is left outside — multiply, then add or subtract.

Picture: the graph of $y = a|x - h| + k$ is a V with its point at (h, k) . The inside hits zero exactly at $x = h$, which is why the V turns there.

Example: For $f(x) = |x - 8| + 1$, find $f(3)$.

Step 1. The input goes inside the bars, so the bars are a grouping symbol here — subtract first and take the absolute value before touching the $+1$.

Step 2. $f(3) = |3 - 8| + 1 = |-5| + 1 = 5 + 1$
 $\Rightarrow f(3) = 6$

Try it: For $g(x) = 3|x + 1| - 4$, find $g(-6)$.

Check: $g(-6) = 3|-6 + 1| - 4 = 3|-5| - 4 = 3(5) - 4 = 15 - 4 = 11$

Skill 2 — Choosing the branch of a piecewise rule

A piecewise rule is a set of formulas plus a set of conditions. The conditions come first.

- Ask: which condition does my input make *true*? Exactly one will.
- Only then substitute — into that branch alone.
- At a boundary, the branch whose sign is $\leq$ or $\geq$ is the one that owns the number. Strict $<$ and $>$ do not.

Example: For $p(x) = 2x - 5$ when $x < 3$ and $p(x) = x^2 + 1$ when $x \geq 3$, find $p(3)$.

Step 1. Test the input against the conditions before substituting: $3 < 3$ is false, and $3 \geq 3$ is true, so the second branch owns this input.

Step 2. $p(3) = 3^2 + 1 = 9 + 1$
 $\Rightarrow p(3) = 10$

Try it: For $q(x) = 4 - x$ when $x < 1$ and $q(x) = 2x^2$ when $x \geq 1$, find $q(1)$.

Check: $q(1) = 2(1)^2 = 2$

Skill 3 — Reading a piecewise model in context

Real prices and fees change rule partway. A flat charge up to some point, then a per-unit charge after it, is a two-branch model: the second branch usually reads $\text{base} + \text{rate} \times (\text{amount past the cutoff})$.

Watch the “past the cutoff” part: it is $w - 2$, not w , when the extra charges start after 2 units. Substituting w there charges the customer twice for the base.

Example: Shipping costs \$5 for a parcel up to 2 kg, and \$5 plus \$2 for each kilogram over 2 kg after that. Find the cost of a 6 kg parcel.

Step 1. Six is past the two-kilogram cutoff, so the second branch applies, and only the weight *above* the cutoff gets charged the per-kilogram rate.

Step 2. $\text{cost} = 5 + 2(6 - 2) = 5 + 2(4) = 5 + 8$

$\Rightarrow$ **13** dollars

Try it: A taxi charges \$7 for the first 3 km and \$3 for each kilometre after that. Find the fare for a 9 km trip.

check: 25 dollars

(!) Watch out: the two errors that cost the most marks are the same error twice. *Substituting before reading* — picking a branch by where it sits on the page instead of by its condition — and *dropping the bars*, so a negative inside stays negative. Both come from starting the arithmetic too early. Read the condition, or close the bars, and only then compute.